STABLE PHASE RETRIEVAL FOR SPANS OF INDEPENDENT RANDOM VARIABLES

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Abstract. We prove that, after L^2 normalization, stable phase retrieval holds over the L^2-spans of independent real-valued random variables if and only if all but possibly one coordinate satisfies a uniform two-sided L^1 bound. This provides a complete characterization of stable phase retrieval for such spaces, building upon the pioneering work by Calderbank–Daubechies–Freeman–Freeman and answering the main question posed by those authors shortly after the completion of their paper.

We provide two different proofs of this fact. The first is a compactness proof based on a probe-function-driven profile decomposition, coupled with the inherent infinite divisibility of limit laws of diffuse sums. The second is a quantitative proof, which uses the head-tail decomposition as a template as in the first proof, but replaces the compactness step by an explicit dichotomy through anticoncentration estimates à la Sperner for diffuse tails. This latter proof was LLM generated based on the ideas in the first proof and a human-based insistence that all steps be quantitative. It was then autoformalized in Lean 4.

1. Introduction

1.1. **Historical background.** In many inverse problems, the phase is precisely the part of the signal that the measurement device cannot detect. This phenomenon is already present in the Pauli problem from quantum mechanics, where one asks how much of a wave function is encoded by the magnitudes of the signal and of its Fourier transform [**?**, **?**, **?**, **?**]; it also appears throughout optics and signal processing, where detectors typically only record (phaseless) intensities rather than amplitudes [**?**, **?**, **?**, **?**]. The mathematical formulation is the phase retrieval problem: given the data, recover the signal up to an ambiguity due to the phase. In the real-valued setting, the unavoidable ambiguity is multiplication by a global sign. A natural question is whether this is the only obstruction to recover the signal.

A substantial finite-dimensional theory has grown around this question for random or structured measurements, including injectivity criteria, stability estimates, and polynomial-time algorithms via convex relaxations [**?**, **?**, **?**, **?**, **?**, **?**, **?**]. The infinite-dimensional theory is more delicate. Phase retrieval can hold in many Hilbert spaces [**?**, **?**], but the passage from uniqueness to a uniform inverse estimate is tricky: uniform stability always fails for continuous frames over infinite dimensional spaces [**?**], and even canonical structured systems such as Gabor measurements can be ill-posed [**?**, **?**, **?**, **?**]. For general background we refer the reader to [**?**]. Further landmarks include early reconstruction and finite-dimensional stability results [**?**, **?**, **?**, **?**, **?**, **?**, **?**, **?**], variants for affine,

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vector-valued, projection-based models or related nonlinear recovery problems [**?**, **?**, **?**, **?**, **?**, **?**, **?**], and infinite-dimensional or structured-measurement results [**?**, **?**, **?**]. For the surrounding frame-theoretic, function space and discretization background we also refer to [**?**, **?**, **?**, **?**, **?**, **?**, **?**, **?**, **?**].

More concretely, two main themes run through the subject. The first is uniqueness: does the map $f \mapsto |f|$ distinguish points of the model space modulo the natural symmetry? The second is stability: can one recover the signal in a uniformly continuous, or even Lipschitz, fashion from phaseless data? While both questions are of great importance in applications, the second one is indispensable for implementation: exact injectivity without stability is too weak for numerical reconstruction, and many infinite-dimensional models exhibit precisely this pathology. One of the main conceptual contributions of [**?**, **?**, **?**, **?**, **?**, **?**] is to make the stability question intrinsic to the geometry of the subspace rather than to a particular discretized frame. This point of view is particularly well suited to random subspaces of L^2, where the ambient structure is probabilistic from the outset and where orthogonality interacts naturally with independence.

We adopt the function-space viewpoint introduced in [**?**, **?**]. Let $(X, \|\cdot\|_X)$ be a real normed function space and let $E \subset X$ be a subspace. We say that E does *stable phase retrieval* if there exists $C < \infty$ such that

$$\min_{\sigma\in\{\pm1\}} \|f - \sigma g\|_X \le C \, \||f| - |g|\|_X \qquad \text{for all } f, g \in E. \tag{1.1}$$

1.2. **Main result.** This paper is concerned with the quantitative aspect of stable phase retrieval, by studying random subspaces E spanned by independent random variables in L^2. Such subspaces are a natural probabilistic analogue of coordinate subspaces, and they sit exactly at the interface between phase retrieval and the geometry of independent sums.

In this setting, Calderbank, Daubechies, Freeman, and Freeman [**?**] proved stable phase retrieval for a shifted simplified model, with the latter two authors proposing to us the following main conjecture regarding the characterization of independent sequences doing stable phase retrieval: if (ξ_i) is an orthonormal sequence of independent centered random variables in L^2, then $\overline{\mathrm{span}}(\xi_i)$ does stable phase retrieval if and only if there exist constants $0 < A \le B < 1$ such that $A \le \|\xi_i\|_{L^1} \le B$ for all but at most one index. Note that the lower bound excludes one from constructing an almost disjoint sequence in $\overline{\mathrm{span}}(\xi_i)$; the upper bound excludes the Rademacher regime in which $|\xi_i|$ is constant and therefore carries no sign information.

The phrase "at most one exceptional coordinate" is not incidental. It is the largest defect compatible with the expected geometry of the problem. One exceptional coordinate may coexist with an otherwise phase-retrieving family, but two genuinely exceptional coordinates are enough to prevent stable phase retrieval to hold.

The necessity of the above conditions is known in the literature: the equivalence of the L^1 and L^2 norms on a stable phase-retrieving subspace follows from [**?**]. Likewise, a centered L^2-normalized variable satisfies $\|\xi\|_{L^1} = 1$ if and only if $\xi^2 \equiv 1$, that is, if and only if ξ is Rademacher-valued. However, a span containing two independent Rademacher random variables cannot even do phase retrieval, since all such coordinates have the same absolute value. The core question is whether these conditions are sufficient. Christ, Pineau and the last-named author [**?**] proved that they suffice under an additional assumption of finite fourth moment.

The main theorem of this manuscript gives the desired sufficiency result, answering completely the central conjecture in this line and providing an explicit, exact characterization of when spans of independent, centered, real-valued L^2-random variables do stable phase retrieval.

Theorem 1.1 (One-exception stable phase retrieval)**.** *Let $\eta, \xi_2, \xi_3, \dots$ be independent real-valued random variables and fix $\delta > 0$. Assume*

$$\|\eta\|_{L^2} = 1, \qquad \mathbb{E}[\eta] = 0,$$

and, for every $i \geq 2$,

$$\|\xi_i\|_{L^2} = 1, \qquad \mathbb{E}[\xi_i] = 0, \qquad \delta \leq \|\xi_i\|_{L^1} \leq 1 - \delta.$$

Then there exists $C_{\delta,\eta} \geq 1$ such that, for all $X, Y \in \overline{\operatorname{span}}(\eta, \xi_i : i \geq 2)$,

$$\min_{\sigma \in \{\pm 1\}} \|X - \sigma Y\|_{L^2} \leq C_{\delta,\eta} \, \||X| - |Y|\|_{L^2}.$$

A theorem of Freeman, Oikhberg, Pineau, and the last-named author [**?**] shows that in $L^2(\mu)$, the problem reduces to orthogonal pairs: to prove stable phase retrieval on a subspace it suffices to lower-bound the modulus gap for pairs of orthogonal unit vectors in that subspace. This orthogonal reduction is the common point of departure of both proofs given here.

Next, we present two proofs of this result. The first proof is qualitative: we assume, for the sake of a contradiction, that the result does not hold. This induces a natural splitting of our random variable: by looking at the coefficients of counterexample-generating pairs of random variables $(\{X_n\}_n, \{Y_n\}_n)$ in the span of $\{\eta, \xi_2, \xi_3, \dots\}$, we may, upon reordering the entries, arrange them so that one obtains first a block which converges strongly. This block is called the 'head' of the sequences. The remaining pieces may be chosen so that they are diffuse, i.e., such that the ℓ^∞-norms of the coefficients are small – hence constituting the 'tails'.

Employing classical characterizations of weak limit of triangular arrays, we are able to derive from counterexample-generating pairs of random variables $(\{X_n\}_n, \{Y_n\}_n)$ an equality of the form $|H_X + R_X| = |H_Y + R_Y|$, where $H_X = \tau H_Y$ for some universal sign $\tau \in \{\pm 1\}$, with (H_X, H_Y) independent of (R_X, R_Y), and the latter random vector being infinitely divisible. From that point on it is relatively straightforward to reach a contradiction by conditioning and using the infinite divisibility of the pair (R_X, R_Y).

The second proof focus on quantitative estimates for the constant $C_{\delta,\eta}$ in Theorem 1.1. It keeps the same head–tail decomposition of the first proof, but it replaces the limit-law endpoint by an explicit head–tail mass-concentration dichotomy, and by anticoncentration estimates for the diffuse part, which acts as a quantitative version of the fact that infinitely divisible laws cannot concentrate on two lines, unless they are actually concentrated near a single line – a crucial feature of the first proof. In the latter regime one partitions the tail into many blocks of comparable energy and applies symmetrization, Sperner-type combinatorics, and small-ball estimates to prove that both combinations stay away from zero with positive probability, thus finishing the proof.

In that way, the compactness and quantitative proofs address the same theorem from complementary angles: the compactness proof shows that a bad sequence cannot exist by passing to a limit model; the quantitative proof shows directly that every orthogonal pair already contains a definite amount of modulus separation.

The paper is organized as follows. Section 2 collects the basic definitions and the analytic tools used throughout. In Section 3 we present the compactness proof of Theorem 1.1, including a Gaussian alternative to the final line-support argument. Section 4 gives a second proof of Theorem 1.1, explaining how the limit-law step can be bypassed by an explicit decomposition together with anticoncentration. Finally, Section 5 discusses extensions, related directions, and open problems.

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LLM Usage

The ideas in the compactness proof were formulated by the authors without the assistance of LLMs. However, after the completion of the core ideas, LLMs played a significant role in the paper. .....

2. Preliminaries

2.1. **Stable phase retrieval and the orthogonal reduction.** We start by recalling the definition of stable phase retrieval in Equation 1.1. Next, we state an important reduction from [**?**, Theorem 3.9] or [**?**, Theorem 2.1] in the simplified setting of L^2 spaces which is our focus here.

Proposition 2.1 (Orthogonal reduction)**.** *Let $E \subset L^2(\mu;\mathbb{R})$ be a subspace. Then E does stable phase retrieval if and only if there exists $c > 0$ such that*

$$\||f| - |g|\|_{L^2} \geq c$$

for every orthogonal pair $f, g \in E$ with $\|f\|_{L^2} = \|g\|_{L^2} = 1$.

Proof. The result from either [**?**, Theorem 3.9] or [**?**, Theorem 2.1] shows that, in order to verify stable phase retrieval, it suffices to only verify it for orthogonal functions f and g such that $\|f\|_{L^2} = 1$ and $\|g\|_{L^2} \leq 1$. Suppose then that the assertion of Proposition 2.1 holds, but stable phase retrieval does *not* hold. Thus, there exists $\{(f_n, g_n)\}_{n\in\mathbb{N}}$ a sequence of pairs satisfying $\|f_n\|_{L^2} = 1 \geq \|g_n\|_{L^2}$ and $f_n \perp g_n$ such that

$$\limsup_{n\to\infty} \||f_n| - |g_n|\|_{L^2} = 0.$$

We first claim that $\|g_n\|_{L^2} \to 1$. Indeed, if not, then passing to a subsequence we would be able to estimate

$$\||f_n| - |g_n|\|_{L^2} \geq \|f_n\|_{L^2} - \|g_n\|_{L^2} \geq c,$$

which contradicts the fact that the pair (f_n, g_n) violates stable phase retrieval at infinity. Next,

$$\||f_n| - |g_n|\|_{L^2} \geq \||f_n| - \lambda_n|g_n|\|_{L^2} - \frac{|\lambda_n - 1|}{\lambda_n}, \tag{2.1}$$

where $\lambda_n := \frac{1}{\|g_n\|_{L^2}}$. By hypothesis, the first summand on the right-hand side of (2.1) is bounded from below by c, while the term $\frac{|\lambda_n-1|}{\lambda_n}$ vanishes because $\|g_n\|_{L^2}$ converges to one. Hence, we reach again a contradiction, which shows that the reduction we were after was indeed valid. □

2.2. **Probe functions and lower L^1 bounds.** The first input is the existence of bounded dual probes. These are the basic coefficient-extraction devices in our proofs, as it will become gradually clear in what follows.

Lemma 2.2 (Linear and quadratic probes)**.** *Let ξ be a real random variable with $\mathbb{E}[\xi] = 0$ and $\mathbb{E}[\xi^2] = 1$.*

(1) *If $\mathbb{E}[|\xi|] \geq A$ for some constant $A > 0$, then there exists a bounded measurable function S such that*
$$\mathbb{E}[S] = 0, \qquad \mathbb{E}[\xi S] = 1, \qquad \|S\|_{L^\infty} \leq \frac{2}{A}.$$
(2) *If $\mathbb{E}[|\xi|] \leq B$ for some constant $B < 1$, then there exists a bounded measurable function T such that*
$$\mathbb{E}[T] = 0, \qquad \mathbb{E}[(\xi^2 - 1)T] = 1, \qquad \|T\|_{L^\infty} \leq \frac{2}{1-B}.$$

Proof. We first construct the linear probe. Define
$$S_0 := \operatorname{sign}(\xi),$$
with the convention $\operatorname{sign}(0) = 0$. Then S_0 is measurable and satisfies $|S_0| \leq 1$ almost surely. Moreover,
$$\xi S_0 = |\xi| \qquad \text{almost surely,}$$
so
$$\mathbb{E}[\xi S_0] = \mathbb{E}[|\xi|] \geq A > 0.$$
This positivity allows us to normalize. Set
$$S := \frac{S_0 - \mathbb{E}[S_0]}{\mathbb{E}[\xi S_0]}.$$
The function S is measurable and bounded. In addition,
$$\mathbb{E}[S] = \frac{\mathbb{E}[S_0] - \mathbb{E}[S_0]}{\mathbb{E}[\xi S_0]} = 0,$$
and, using that $\mathbb{E}[\xi] = 0$,
$$\mathbb{E}[\xi S] = \frac{\mathbb{E}[\xi S_0] - \mathbb{E}[S_0]\mathbb{E}[\xi]}{\mathbb{E}[\xi S_0]} = \frac{\mathbb{E}[\xi S_0]}{\mathbb{E}[\xi S_0]} = 1.$$
Finally, since $|S_0| \leq 1$ and $|\mathbb{E}[S_0]| \leq \mathbb{E}|S_0| \leq 1$,
$$|S_0 - \mathbb{E}[S_0]| \leq 2 \qquad \text{almost surely,}$$
and therefore
$$\|S\|_{L^\infty} \leq \frac{2}{\mathbb{E}[\xi S_0]} = \frac{2}{\mathbb{E}[|\xi|]} \leq \frac{2}{A}.$$
This proves part (1).

For the quadratic probe, apply the same idea to the centered variable
$$\eta := \xi^2 - 1.$$

Because $\mathbb{E}[\xi^2] = 1$, one has $\mathbb{E}[\eta] = 0$. We now show that η has positive L^1 mass bounded away from zero. Since $\xi^2 = |\xi|^2$,

$$|\eta| = ||\xi|^2 - 1| = ||\xi| - 1|(|\xi| + 1) \geq ||\xi| - 1|.$$

Taking expectations and using the reverse triangle inequality yields

$$\mathbb{E}[|\eta|] \geq \mathbb{E}[||\xi| - 1|] \geq \big|\mathbb{E}[|\xi|] - 1\big| = 1 - \mathbb{E}[|\xi|] \geq 1 - B.$$

Now set

$$T_0 := \operatorname{sign}(\eta)$$

and define

$$T := \frac{T_0 - \mathbb{E}[T_0]}{\mathbb{E}[\eta T_0]} = \frac{T_0 - \mathbb{E}[T_0]}{\mathbb{E}[|\eta|]}.$$

Exactly as in the linear case, T is measurable, bounded, and satisfies

$$\mathbb{E}[T] = 0, \qquad \mathbb{E}[\eta T] = 1.$$

Since $\eta = \xi^2 - 1$, we have

$$\mathbb{E}[(\xi^2 - 1)T] = 1.$$

The L^∞ bound is

$$\|T\|_{L^\infty} \leq \frac{2}{\mathbb{E}[|\eta|]} \leq \frac{2}{1 - B}.$$

This completes the proof. □

Remark 2.3. If $\xi_1, \ldots, \xi_n$ are independent centered random variables and $X_c = \sum_i c_i \xi_i$, then the centering of the probes eliminates every coordinate except the one being tested. In particular,

$$\mathbb{E}[X_c\, S_i] = c_i, \qquad \mathbb{E}[X_c^2\, T_i] = c_i^2, \qquad \mathbb{E}[X_c^2\, S_i S_j] = 2c_i c_j \quad (i \neq j),$$

whenever S_i, S_j, T_i are the corresponding probes for ξ_i, ξ_j.

The second basic estimate allow us to transfer the lower bounds for the L^1 norm of the random variables to its entire spam (up to a different constant). More accurately, we prove the following result.

Lemma 2.4 (L^1 lower bound for independent sums)**.** *Let $\zeta_1, \ldots, \zeta_N$ be independent centered real random variables and assume that*

$$\|\zeta_i\|_{L^1} \geq \delta \|\zeta_i\|_{L^2} \qquad (1 \leq i \leq N).$$

Then for every scalar family $(c_i)_{i=1}^N$,

$$\|\sum_{i=1}^N c_i \zeta_i\|_{L^1} \geq \frac{\delta}{2\sqrt{2}} \Big(\sum_{i=1}^N c_i^2 \|\zeta_i\|_{L^2}^2 \Big)^{1/2}.$$

Proof. The statement is homogeneous in the coefficient vector, so we may first normalize and assume that

$$\sum_{i=1}^N c_i^2 \|\zeta_i\|_{L^2}^2 = 1.$$

Indeed, once the claim is established under this normalization, the general case follows by scaling.

We next reduce to the unit-variance case. For each i set

$$\widetilde{\zeta}_i := \frac{\zeta_i}{\|\zeta_i\|_{L^2}}, \qquad \widetilde{c}_i := c_i\|\zeta_i\|_{L^2}.$$

The variables $\widetilde{\zeta}_i$ are still independent and centered, they satisfy $\|\widetilde{\zeta}_i\|_{L^2} = 1$, and the lower L^1 bound becomes

$$\|\widetilde{\zeta}_i\|_{L^1} = \frac{\|\zeta_i\|_{L^1}}{\|\zeta_i\|_{L^2}} \geq \delta.$$

Moreover,

$$\sum_{i=1}^N \widetilde{c}_i\widetilde{\zeta}_i = \sum_{i=1}^N c_i\zeta_i, \qquad \sum_{i=1}^N \widetilde{c}_i^2 = 1.$$

Thus it is enough to prove the lemma under the additional assumptions

$$\|\zeta_i\|_{L^2} = 1 \qquad (1 \leq i \leq N), \qquad \sum_{i=1}^N c_i^2 = 1.$$

Define

$$S := \sum_{i=1}^N c_i\zeta_i,$$

and let

$$S' := \sum_{i=1}^N c_i\zeta_i'$$

be an independent copy of S, obtained from an independent copy $(\zeta_i')_{i=1}^N$ of the family $(\zeta_i)_{i=1}^N$. By the triangle inequality,

$$|S - S'| \leq |S| + |S'|.$$

Taking expectations and using that S and S' have the same distribution, we obtain

$$\|S - S'\|_{L^1} \leq 2\|S\|_{L^1}.$$

Equivalently,

$$2\|S\|_{L^1} \geq \|S - S'\|_{L^1}.$$

Now the random variables $\zeta_i - \zeta_i'$ are independent and symmetric. Conditioning on their absolute values and applying the Khintchine inequality gives

$$\|S - S'\|_{L^1} = \Big\|\sum_{i=1}^N c_i(\zeta_i - \zeta_i')\Big\|_{L^1} \geq \frac{1}{\sqrt{2}}\,\mathbb{E}\Big[\Big(\sum_{i=1}^N c_i^2(\zeta_i - \zeta_i')^2\Big)^{1/2}\Big].$$

Because the nonnegative numbers $(c_i^2)_{i=1}^N$ sum to one, Cauchy–Schwarz implies the pointwise estimate

$$\sum_{i=1}^N c_i^2|\zeta_i - \zeta_i'| \leq \Big(\sum_{i=1}^N c_i^2\Big)^{1/2}\Big(\sum_{i=1}^N c_i^2(\zeta_i - \zeta_i')^2\Big)^{1/2} = \Big(\sum_{i=1}^N c_i^2(\zeta_i - \zeta_i')^2\Big)^{1/2}.$$

Taking expectations and combining with the previous inequality, we arrive at

$$\|S - S'\|_{L^1} \geq \frac{1}{\sqrt{2}}\sum_{i=1}^N c_i^2\|\zeta_i - \zeta_i'\|_{L^1}.$$

Hence

$$\|S\|_{L^1} \geq \frac{1}{2\sqrt{2}} \sum_{i=1}^{N} c_i^2 \|\zeta_i - \zeta_i'\|_{L^1}.$$

It remains to bound each $\|\zeta_i - \zeta_i'\|_{L^1}$ from below. Since ζ_i' is independent of ζ_i and centered,

$$\mathbb{E}[\zeta_i - \zeta_i' \mid \zeta_i] = \zeta_i.$$

Applying Jensen's inequality to the conditional expectation gives

$$\|\zeta_i - \zeta_i'\|_{L^1} \geq \|\mathbb{E}[\zeta_i - \zeta_i' \mid \zeta_i]\|_{L^1} = \|\zeta_i\|_{L^1} \geq \delta.$$

Substituting this into the previous bound yields

$$\|S\|_{L^1} \geq \frac{\delta}{2\sqrt{2}} \sum_{i=1}^{N} c_i^2 = \frac{\delta}{2\sqrt{2}}.$$

Since this is exactly the normalized form of the desired estimate, the proof is complete. □

2.3. **Diffusive arrays and infinitely divisible limits.** A key component in our proof is the rigidity of the weak limits, the next lemma is a precise statement of this type.

Lemma 2.5 (Diffuse tail rigidity)**.** *Let $(\xi_{n,i})_{n\geq 1,\ i\geq 2}$ be a triangular array such that for each fixed n the family $(\xi_{n,i})_{i\geq 2}$ is independent, centered, L^2-normalized, and satisfies*

$$\|\xi_{n,i}\|_{L^1} \geq \delta \qquad (i \geq 2).$$

Let $c^{(n)} \in \ell^2(\mathbb{N})$ satisfy

$$\sup_n \|c^{(n)}\|_2 < \infty, \qquad c_1^{(n)} = 0, \qquad \max_{i\geq 2} |c_i^{(n)}| \to 0.$$

Set

$$Z_n := \sum_{i\geq 2} c_i^{(n)} \xi_{n,i}.$$

If $Z_n \to 0$ in distribution, then $\|c^{(n)}\|_2 \to 0$.

Proof. Because the limiting law is the Dirac mass at 0, the assumption $Z_n \to 0$ in distribution implies that

$$Z_n \to 0 \qquad \text{in probability.}$$

To pass from convergence in probability to convergence of first moments, we use uniform integrability.

For each n, the variables $(\xi_{n,i})_{i\geq 2}$ are independent, centered, and L^2-normalized. Therefore they form an orthonormal family in L^2, and the random sum Z_n satisfies

$$\|Z_n\|_{L^2}^2 = \mathbb{E}\Big[\Big(\sum_{i\geq 2} c_i^{(n)} \xi_{n,i}\Big)^2\Big] = \sum_{i\geq 2} (c_i^{(n)})^2 \mathbb{E}[\xi_{n,i}^2] = \sum_{i\geq 2} (c_i^{(n)})^2 = \|c^{(n)}\|_2^2.$$

Since $\sup_n \|c^{(n)}\|_2 < \infty$, the sequence $(Z_n)_n$ is bounded in L^2. Any L^2-bounded family is uniformly integrable in L^1, so $(|Z_n|)_n$ is uniformly integrable. Because $Z_n \to 0$ in probability, we may therefore conclude that

$$\|Z_n\|_{L^1} \longrightarrow 0.$$

We now apply Lemma 2.4 to each row of the array. The assumptions of that lemma are satisfied because every $\xi_{n,i}$ is centered, L^2-normalized, and obeys $\|\xi_{n,i}\|_{L^1} \geq \delta$. Hence

$$\|Z_n\|_{L^1} = \Big\|\sum_{i\geq 2} c_i^{(n)}\xi_{n,i}\Big\|_{L^1} \geq \frac{\delta}{2\sqrt{2}}\Big(\sum_{i\geq 2}(c_i^{(n)})^2\Big)^{1/2} = \frac{\delta}{2\sqrt{2}}\|c^{(n)}\|_2.$$

The left-hand side tends to zero, so the right-hand side must also tend to zero. Therefore

$$\|c^{(n)}\|_2 \longrightarrow 0,$$

which is the desired conclusion. □

In order to state our next result, we recall that a $\mathbb{R}^n$-valued random variable S is *infinitely divisible* if, for each positive integer n, there exist i.i.d. random variables $\eta_{n,1}, \eta_{n,2}, \ldots, \eta_{n,n}$ such that

$$S \overset{d}{\sim} \eta_{n,1} + \cdots + \eta_{n,n}.$$

Infinite divisible distributions are a fundamental object in probability theory due to the classical of limit theorem of triangular arrays. See for instance, [**?**, Chapter IV]. The next result shows that the weak limit of diffuse random vectors is an infinitely divisible random variable.

Lemma 2.6 (Diffuse limits are infinitely divisible)**.** *Let $d \geq 1$, and let $(\eta_{n,i})_{n\geq 1,\ i\geq 1}$ be a triangular array of $\mathbb{R}^d$-valued random variables. Assume that, for each fixed n, the variables $(\eta_{n,i})_{i\geq 1}$ are independent and that the row sum*

$$S_n := \sum_{i\geq 1}\eta_{n,i}$$

is well defined as a limit in probability of its finite partial sums. Assume also that the array is infinitesimal, in the sense that for every $\varepsilon > 0$,

$$\sup_{i\geq 1}\mathbb{P}\{\|\eta_{n,i}\| > \varepsilon\} \longrightarrow 0.$$

If S_n converges in distribution to an $\mathbb{R}^d$-valued random variable S, then the law of S is infinitely divisible.

Proof. The finite-row version is classical: if $U_{n,1}, \ldots, U_{n,k_n}$ are independent $\mathbb{R}^d$-valued random variables,

$$\max_{1\leq j\leq k_n}\mathbb{P}\{\|U_{n,j}\| > \varepsilon\} \longrightarrow 0 \qquad (\varepsilon > 0),$$

and

$$\sum_{j=1}^{k_n} U_{n,j} \Rightarrow U,$$

then the law of U is infinitely divisible. This is the classical Kolmogorov–Khintchine theorem for null arrays; see [**?**, Theorem 9.3]. In the one-dimensional case it is also the classical theorem of Gnedenko–Kolmogorov [**?**, Chapter IV, Section 24].

It remains only to reduce the present countable-row formulation to that finite-row theorem. Since S_n is the limit in probability of its finite partial sums, for each n we may choose an integer k_n such

that

$$\mathbb{P}\left\{\|S_n - \sum_{i=1}^{k_n} \eta_{n,i}\| > \frac{1}{n}\right\} \leq \frac{1}{n}.$$

Set

$$S_n^{(k)} := \sum_{i=1}^{k_n} \eta_{n,i}.$$

Then $S_n^{(k)} - S_n \to 0$ in probability and hence, by Slutsky's theorem,

$$S_n^{(k)} = (S_n^{(k)} - S_n) + S_n \Rightarrow S.$$

For each fixed n, the summands $\eta_{n,1}, \ldots, \eta_{n,k_n}$ are independent, and the finite array remains infinitesimal because

$$\max_{1\leq i\leq k_n} \mathbb{P}\{\|\eta_{n,i}\| > \varepsilon\} \leq \sup_{i\geq 1} \mathbb{P}\{\|\eta_{n,i}\| > \varepsilon\} \longrightarrow 0.$$

Applying the finite-row theorem gives that the law of S is infinitely divisible. □

3. First Proof: Compactness

This section proves Theorem 1.1 through the first method mentioned in the introduction, consisting of an argument by contradiction through extracting limit properties of sequences failing SPR.

3.1. Contradiction array and profile decomposition.

We start with the basic setup that failure to satisfy SPR gives.

Proposition 3.1 (Normalized counterexample array)**.** *If the conclusion of Theorem 1.1 is false, then there exist*

(i) *a triangular array $(\xi_{n,i})_{n\geq 1,\ i\geq 2}$ such that for each fixed n the family $(\eta, \xi_{n,i})_{i\geq 2}$ is independent and every $\xi_{n,i}$ satisfies*

$$\|\xi_{n,i}\|_{L^2} = 1, \qquad \mathbb{E}[\xi_{n,i}] = 0, \qquad \delta \leq \|\xi_{n,i}\|_{L^1} \leq 1 - \delta;$$

(ii) *coefficient vectors $a^{(n)}, b^{(n)} \in \ell^2(\mathbb{N})$ with*

$$\|a^{(n)}\|_2 = \|b^{(n)}\|_2 = 1, \qquad \langle a^{(n)}, b^{(n)}\rangle = 0,$$

such that, writing

$$X_n := a_1^{(n)}\eta + \sum_{i\geq 2} a_i^{(n)}\xi_{n,i}, \qquad Y_n := b_1^{(n)}\eta + \sum_{i\geq 2} b_i^{(n)}\xi_{n,i},$$

one has

$$\||X_n| - |Y_n|\|_{L^2} \longrightarrow 0.$$

Proof. Assume that the conclusion of Theorem 1.1 is false. Then there is no constant C for which the stable phase retrieval estimate holds uniformly on the closed span of the coordinates. By the orthogonal reduction (Proposition 2.1), this means that we may find, for each $n \in \mathbb{N}$, an orthogonal pair of unit vectors in the span whose modulus gap is at most $1/n$. Concretely, there exist functions

X_n, Y_n in $\overline{\mathrm{span}}(\eta, \xi_i : i \geq 2)$ such that

$$\|X_n\|_{L^2} = \|Y_n\|_{L^2} = 1, \qquad \mathbb{E}[X_n Y_n] = 0, \qquad \||X_n| - |Y_n|\|_{L^2} \leq \frac{1}{n}.$$

Because the family $(\eta, \xi_i)_{i\geq 2}$ is orthonormal in L^2, we may write

$$X_n = a_1^{(n)}\eta + \sum_{i\geq 2} a_i^{(n)}\xi_i, \qquad Y_n = b_1^{(n)}\eta + \sum_{i\geq 2} b_i^{(n)}\xi_i,$$

with coefficient vectors $a^{(n)}, b^{(n)} \in \ell^2(\mathbb{N})$. Parseval's identity then yields

$$\|a^{(n)}\|_2 = \|X_n\|_{L^2} = 1, \qquad \|b^{(n)}\|_2 = \|Y_n\|_{L^2} = 1,$$

and, since the two functions are orthogonal in L^2,

$$\langle a^{(n)}, b^{(n)}\rangle = \mathbb{E}[X_n Y_n] = 0.$$

Finally, to prepare for later rowwise permutations of the good coordinates (the ones with L^1 norm bounded away from one), we rename the good variables used in the n-th row by writing them as $(\xi_{n,i})_{i\geq 2}$. This does not alter their laws or any of the stated hypotheses, but it is convenient for the compactness scheme. The displayed modulus convergence is exactly our chosen violating sequence. This proves the proposition. □

Lemma 3.2 (Reordered profile decomposition)**.** *After passing to a subsequence and, row by row, applying the same permutation to the good coordinates and to their coefficients, one may find vectors $a, b \in \ell^2(\mathbb{N})$ and integers $m_n \to \infty$ such that, with*

$$a_{\mathrm{head}}^{(n)} := a^{(n)}\mathbf{1}_{\{1,\dots,m_n\}}, \qquad a_{\mathrm{tail}}^{(n)} := a^{(n)}\mathbf{1}_{\{m_n+1,m_n+2,\dots\}},$$

and likewise for $b^{(n)}$, the following hold:

(1) $a_{\mathrm{head}}^{(n)} \to a$ *and* $b_{\mathrm{head}}^{(n)} \to b$ *strongly in* ℓ^2*;*

(2)
$$\max_{i>m_n} \max(|a_i^{(n)}|, |b_i^{(n)}|) \to 0;$$

(3)
$$\|a_{\mathrm{tail}}^{(n)}\|_2^2 \to 1 - \|a\|_2^2, \qquad \|b_{\mathrm{tail}}^{(n)}\|_2^2 \to 1 - \|b\|_2^2,$$

and

$$\langle a_{\mathrm{tail}}^{(n)}, b_{\mathrm{tail}}^{(n)}\rangle \to -\langle a, b\rangle.$$

Proof. For each row n and each good index $i \geq 2$, define

$$s_i^{(n)} := \max\big(|a_i^{(n)}|, |b_i^{(n)}|\big).$$

Permuting the good coordinates within the n-th row, we may assume that the sequence $(s_i^{(n)})_{i\geq 2}$ is nonincreasing in i. The same permutation is applied simultaneously to the variables $\xi_{n,i}$ and to both coefficient vectors, so our hypothesis is not changed.

The monotone rearrangement gives a uniform pointwise bound on the reordered coefficients. Indeed,

$$\sum_{i\geq 2}(s_i^{(n)})^2 \leq \sum_{i\geq 2}\big((a_i^{(n)})^2 + (b_i^{(n)})^2\big) \leq 2.$$

Since $(s_i^{(n)})_{i\geq 2}$ is nonincreasing, for every $i \geq 2$ we have

$$(i-1)(s_i^{(n)})^2 \leq \sum_{j=2}^{i} (s_j^{(n)})^2 \leq 2,$$

and therefore

$$s_i^{(n)} \leq \sqrt{\frac{2}{i-1}}.$$

In particular, for each fixed coordinate i, the sequences $\left(a_i^{(n)}\right)_{n\geq 1}$ and $\left(b_i^{(n)}\right)_{n\geq 1}$ are bounded.

We now extract coordinatewise limits. First pass to a subsequence so that $a_1^{(n)}$ and $b_1^{(n)}$ converge. Next pass to a further subsequence so that $a_2^{(n)}$ and $b_2^{(n)}$ converge, and continue inductively. A standard diagonal argument yields a single subsequence, not relabeled, for which every fixed coordinate converges:

$$a_i^{(n)} \to a_i, \qquad b_i^{(n)} \to b_i \qquad (i \geq 1).$$

Set

$$a := (a_i)_{i\geq 1}, \qquad b := (b_i)_{i\geq 1}.$$

We claim that $a, b \in \ell^2(\mathbb{N})$. Fix $M \in \mathbb{N}$. Since the first M coordinates converge,

$$\sum_{i=1}^{M} a_i^2 = \lim_{n\to\infty} \sum_{i=1}^{M} (a_i^{(n)})^2 \leq \liminf_{n\to\infty} \sum_{i\geq 1} (a_i^{(n)})^2 = 1.$$

Letting $M \to \infty$ and using monotone convergence gives $\sum_i a_i^2 \leq 1$. The same argument yields $\sum_i b_i^2 \leq 1$.

Because $a, b \in \ell^2$, we may choose an increasing sequence of integers $(m_k)_{k\geq 1}$ with $m_k \to \infty$ such that

$$\sum_{i>m_k} a_i^2 + \sum_{i>m_k} b_i^2 \leq 2^{-k} \qquad (k \geq 1).$$

For each fixed k, coordinatewise convergence implies

$$\sum_{i=1}^{m_k} |a_i^{(n)} - a_i|^2 + \sum_{i=1}^{m_k} |b_i^{(n)} - b_i|^2 \longrightarrow 0 \qquad (n \to \infty).$$

Passing to another subsequence if necessary, we may assume that for each n,

$$\sum_{i=1}^{m_n} |a_i^{(n)} - a_i|^2 + \sum_{i=1}^{m_n} |b_i^{(n)} - b_i|^2 \leq 2^{-n}.$$

We now verify the three conclusions. First,

$$\|a_{\text{head}}^{(n)} - a\|_2^2 \leq \sum_{i=1}^{m_n} |a_i^{(n)} - a_i|^2 + \sum_{i>m_n} a_i^2 \leq 2^{-n} + 2^{-n},$$

so $a_{\text{head}}^{(n)} \to a$ in ℓ^2. The same estimate gives $b_{\text{head}}^{(n)} \to b$.

Second, if $i > m_n$, then by monotonicity,

$$\max\left(|a_i^{(n)}|, |b_i^{(n)}|\right) = s_i^{(n)} \leq s_{m_n+1}^{(n)} \leq \sqrt{\frac{2}{m_n}}.$$

Since $m_n \to \infty$, this proves

$$\max_{i>m_n} \max\big(|a_i^{(n)}|, |b_i^{(n)}|\big) \to 0.$$

Finally, because each full coefficient vector has ℓ^2 norm one,

$$\|a_{\text{tail}}^{(n)}\|_2^2 = 1 - \|a_{\text{head}}^{(n)}\|_2^2 \longrightarrow 1 - \|a\|_2^2,$$

and similarly

$$\|b_{\text{tail}}^{(n)}\|_2^2 \longrightarrow 1 - \|b\|_2^2.$$

For the inner products, use the orthogonality of the full vectors:

$$0 = \langle a^{(n)}, b^{(n)} \rangle = \langle a_{\text{head}}^{(n)}, b_{\text{head}}^{(n)} \rangle + \langle a_{\text{tail}}^{(n)}, b_{\text{tail}}^{(n)} \rangle.$$

Since the head component converges strongly,

$$\langle a_{\text{head}}^{(n)}, b_{\text{head}}^{(n)} \rangle \to \langle a, b \rangle,$$

and therefore

$$\langle a_{\text{tail}}^{(n)}, b_{\text{tail}}^{(n)} \rangle \to -\langle a, b \rangle.$$

This proves the lemma. □

3.2. **Head identification.** Our next step is to identify the behavior of the head profiles constructed in the previous step when n is sufficiently large. To this end, we start with an analogue of Lemma 2.2 for the exceptional coordinate η.

Lemma 3.3 (Exceptional probes)**.** *There exists a bounded measurable function S_η such that*

$$\mathbb{E}[S_\eta] = 0, \qquad \mathbb{E}[\eta S_\eta] = 1.$$

If $\eta^2 \not\equiv 1$ almost surely, then there exists a bounded measurable function T_η such that

$$\mathbb{E}[T_\eta] = 0, \qquad \mathbb{E}[(\eta^2 - 1)T_\eta] = 1.$$

Proof. Since $\|\eta\|_{L^2} = 1$, the random variable η is not almost surely zero. Hence $\mathbb{E}[|\eta|] > 0$. Applying the linear-probe construction from Lemma 2.2 with $A = \mathbb{E}[|\eta|]$ produces a bounded measurable function S_η satisfying

$$\mathbb{E}[S_\eta] = 0, \qquad \mathbb{E}[\eta S_\eta] = 1.$$

Assume now that $\eta^2 \not\equiv 1$ almost surely. Then the centered random variable $\eta^2 - 1$ is not almost surely zero, because

$$\mathbb{E}[\eta^2 - 1] = \mathbb{E}[\eta^2] - 1 = 0.$$

Therefore its L^1 norm is strictly positive:

$$\mathbb{E}[|\eta^2 - 1|] > 0.$$

Applying again the linear-probe construction, now to the random variable $\eta^2 - 1$, we obtain a bounded measurable function T_η such that

$$\mathbb{E}[T_\eta] = 0, \qquad \mathbb{E}[(\eta^2 - 1)T_\eta] = 1.$$

This is exactly the desired conclusion. □

Lemma 3.4 (Identification of the head profile)**.** *Either there exists $\tau \in \{\pm 1\}$ such that*

$$a = \tau b \qquad \text{and} \qquad \|a^{(n)}_{\text{head}} - \tau b^{(n)}_{\text{head}}\|_2 \to 0,$$

or else η is a Rademacher variable and

$$a_i = b_i = 0 \qquad (i \geq 2).$$

Proof. We first convert the convergence of moduli into convergence of the quadratic observable. Since

$$\|X_n\|_{L^2} = \|Y_n\|_{L^2} = 1,$$

we have

$$\||X_n| + |Y_n|\|_{L^2} \leq 2.$$

Therefore

$$\||X_n|^2 - |Y_n|^2\|_{L^1} = \|(|X_n| - |Y_n|)(|X_n| + |Y_n|)\|_{L^1} \leq 2\||X_n| - |Y_n|\|_{L^2} \longrightarrow 0.$$

Because the variables are real-valued, $|X_n|^2 = X_n^2$ and $|Y_n|^2 = Y_n^2$, so we have established

$$\|X_n^2 - Y_n^2\|_{L^1} \longrightarrow 0.$$

Step 1: diagonal good-coordinate identities. Fix $i \geq 2$ and let $T_{n,i}$ be the quadratic probe associated with the coordinate $\xi_{n,i}$. The probes are uniformly bounded in L^∞, so the L^1 convergence above implies

$$\mathbb{E}[(X_n^2 - Y_n^2)T_{n,i}] \longrightarrow 0.$$

We compute the left-hand side explicitly. Expanding X_n^2 and integrating against $T_{n,i}$, every term involving a coordinate different from $\xi_{n,i}$ vanishes because $T_{n,i}$ depends only on $\xi_{n,i}$ and is centered. Mixed terms involving $\xi_{n,i}$ and another coordinate also vanish by independence and centering. The only surviving contribution is the term $(a_i^{(n)})^2\xi_{n,i}^2$. Hence

$$\mathbb{E}[X_n^2 T_{n,i}] = (a_i^{(n)})^2\mathbb{E}[\xi_{n,i}^2 T_{n,i}] = (a_i^{(n)})^2,$$

because

$$\mathbb{E}[\xi_{n,i}^2 T_{n,i}] = \mathbb{E}[(\xi_{n,i}^2 - 1)T_{n,i}] + \mathbb{E}[T_{n,i}] = 1 + 0.$$

Similarly,

$$\mathbb{E}[Y_n^2 T_{n,i}] = (b_i^{(n)})^2.$$

Therefore

$$(a_i^{(n)})^2 - (b_i^{(n)})^2 = \mathbb{E}[(X_n^2 - Y_n^2)T_{n,i}] \longrightarrow 0.$$

Passing to the limit in n gives

$$a_i^2 = b_i^2 \qquad (i \geq 2).$$

Step 2: off-diagonal good-good identities. Fix distinct indices $i, j \geq 2$ and let $S_{n,i}, S_{n,j}$ be the corresponding linear probes. Again boundedness and L^1 convergence imply

$$\mathbb{E}[(X_n^2 - Y_n^2)S_{n,i}S_{n,j}] \longrightarrow 0.$$

In the expansion of X_n^2, all terms vanish after integration against $S_{n,i}S_{n,j}$ except the mixed term $2a_i^{(n)}a_j^{(n)}\xi_{n,i}\xi_{n,j}$. Indeed, any term missing one of the coordinates $\xi_{n,i}$ or $\xi_{n,j}$ is killed by the zero

mean of the corresponding probe, and terms involving other coordinates factor through zero means by independence. Hence

$$\mathbb{E}[X_n^2 S_{n,i} S_{n,j}] = 2a_i^{(n)} a_j^{(n)} \mathbb{E}[\xi_{n,i} S_{n,i}]\, \mathbb{E}[\xi_{n,j} S_{n,j}] = 2a_i^{(n)} a_j^{(n)}.$$

Similarly,

$$\mathbb{E}[Y_n^2 S_{n,i} S_{n,j}] = 2b_i^{(n)} b_j^{(n)}.$$

Thus

$$2(a_i^{(n)} a_j^{(n)} - b_i^{(n)} b_j^{(n)}) \longrightarrow 0,$$

and passing to the limit gives

$$a_i a_j = b_i b_j \qquad (i \neq j,\ i, j \geq 2).$$

Step 3: exceptional-good identities. Fix $j \geq 2$. Let S_η be the bounded probe from Proposition 3.3. By the same reasoning,

$$\mathbb{E}[(X_n^2 - Y_n^2) S_\eta S_{n,j}] \longrightarrow 0.$$

Once more only one mixed term survives, namely $2a_1^{(n)} a_j^{(n)} \eta \xi_{n,j}$ for X_n^2 and the corresponding term for Y_n^2. Therefore

$$\mathbb{E}[X_n^2 S_\eta S_{n,j}] = 2a_1^{(n)} a_j^{(n)} \mathbb{E}[\eta S_\eta] \mathbb{E}[\xi_{n,j} S_{n,j}] = 2a_1^{(n)} a_j^{(n)},$$

and analogously

$$\mathbb{E}[Y_n^2 S_\eta S_{n,j}] = 2b_1^{(n)} b_j^{(n)}.$$

Hence

$$a_1 a_j = b_1 b_j \qquad (j \geq 2).$$

Step 4: deduction of the sign. Assume first that there exists some index $i_0 \geq 2$ such that $a_{i_0} \neq 0$. From the diagonal identity we know that $a_{i_0}^2 = b_{i_0}^2$, so $b_{i_0} = \tau a_{i_0}$ for some $\tau \in \{\pm 1\}$. Then for every $j \geq 2$, the off-diagonal identity gives

$$a_{i_0} a_j = b_{i_0} b_j = \tau a_{i_0} b_j.$$

Since $a_{i_0} \neq 0$, we conclude that $a_j = \tau b_j$ for every $j \geq 2$. Applying the exceptional-good identity with $j = i_0$, we obtain

$$a_1 a_{i_0} = b_1 b_{i_0} = \tau a_{i_0} b_1,$$

and again $a_{i_0} \neq 0$ implies $a_1 = \tau b_1$. Hence $a = \tau b$.

Assume next that

$$a_i = 0 \qquad \text{for every } i \geq 2.$$

Then the diagonal identities show that also $b_i = 0$ for every $i \geq 2$. If $\eta^2 \not\equiv 1$ almost surely, let T_η be the quadratic probe from Lemma 3.3. Repeating the diagonal argument for the exceptional coordinate gives

$$(a_1^{(n)})^2 - (b_1^{(n)})^2 = \mathbb{E}[(X_n^2 - Y_n^2) T_\eta] \longrightarrow 0,$$

whence $a_1^2 = b_1^2$ and therefore $a = \tau b$ for some sign τ.

The only remaining possibility is that $\eta^2 \equiv 1$ almost surely. Since $\mathbb{E}[\eta] = 0$ and $\|\eta\|_{L^2} = 1$, this means exactly that η is a Rademacher variable. In that case we are in the second alternative stated in the lemma.

Finally, if the first alternative holds, then the convergence of the head profiles follows from Lemma 3.2:

$$\|a_{\text{head}}^{(n)} - \tau b_{\text{head}}^{(n)}\|_2 \leq \|a_{\text{head}}^{(n)} - a\|_2 + \|b_{\text{head}}^{(n)} - b\|_2 \longrightarrow 0.$$

This finishes the proof. □

3.3. **Limit law and exact modulus identity.** We next identify the properties that the limiting distributions stemming from sequences failing SPR must satisfy.

Lemma 3.5 (Limit law)**.** *After passing to a subsequence, the quadruple*

$$\left(X_n^{\text{head}}, Y_n^{\text{head}}, X_n^{\text{tail}}, Y_n^{\text{tail}}\right)$$

converges in distribution to a random vector

$$(H_X, H_Y, R_X, R_Y) \in \mathbb{R}^4$$

such that:

(1) (H_X, H_Y) *is independent of* (R_X, R_Y)*;*
(2) *the law of* (R_X, R_Y) *is infinitely divisible;*
(3) *either* $H_X = \tau H_Y$ *almost surely for some* $\tau \in \{\pm 1\}$*, or else* η *is Rademacher and* $H_X = a_1\eta$*,* $H_Y = b_1\eta$ *almost surely;*
(4)
$$|H_X + R_X| = |H_Y + R_Y| \qquad \textit{almost surely.}$$

Proof. The head variables converge in L^2 by the strong convergence of the head coefficients and the orthonormality of each row. The tail coefficient arrays are infinitesimal by Lemma 3.2, so every weak limit of the tail pair is infinitely divisible by Lemma 2.6. Independence of head and tail persists because they are built from disjoint independent coordinates. Finally, $\||X_n| - |Y_n|\|_{L^2} \to 0$ implies $|X_n| - |Y_n| \to 0$ in probability; passing to the joint weak limit gives the exact identity in the limit. The description of the head follows from Lemma 3.4. □

Lemma 3.6 (Infinitely divisible laws on the phase-retrieval cone are line-supported)**.** *Let* μ *be an infinitely divisible probability measure on* $\mathbb{R}^2$ *whose support satisfies*

$$\operatorname{supp}\mu \subset \{(u,v) \in \mathbb{R}^2 : |u| = |v|\} =: \Gamma.$$

Then there exists $\tau \in \{\pm 1\}$ *such that*

$$\operatorname{supp}\mu \subset \{(u, \tau u) : u \in \mathbb{R}\} =: L_\tau.$$

Proof. The set Γ is exactly the union of two lines $L_+ \cup L_-$. Since μ is infinitely divisible, it is in particular 2-divisible: there exists a probability measure ν such that $\mu = \nu * \nu$. Let $A = \operatorname{supp}\nu$. Then

$$\operatorname{supp}\mu = \overline{A + A} \subset L_+ \cup L_-.$$

If $x, y \in A$ are distinct, then $2x, x+y, 2y \in A + A$. If $2x$ and $2y$ belong to the same line L_τ, then so does $x + y$, forcing $x, y \in L_\tau$. If they belong to different lines, then the midpoint $x + y$ is the origin, so $y = -x$ and again x, y lie in one of the lines. Thus every two points of A lie on a common

phase-retrieval line. Since $L_+ \cap L_- = \{0\}$, this implies that A is contained in one of the two lines, and so is $\operatorname{supp}\mu$. □

3.4. **Conclusion of the compactness proof.**

Proof of Theorem 1.1. Assume that Theorem 1.1 fails and let the counterexample array be as in Proposition 3.1. Apply Lemmas 3.2, 3.4, and 3.5.

If the first alternative of Lemma 3.4 holds, fix $\tau \in \{\pm 1\}$ such that $H_Y = \tau H_X$ almost surely. The exact modulus identity becomes

$$|H_X + R_X| = |\tau H_X + R_Y| \qquad \text{almost surely.}$$

If H_X is not almost surely zero, then conditioning on the tail and varying the head over two support points of H_X shows that $R_Y - \tau R_X = 0$ almost surely. If $H_X \equiv 0$, then (R_X, R_Y) itself satisfies $|R_X| = |R_Y|$ almost surely, so Lemma 3.6 yields $R_Y = \tau_0 R_X$ almost surely for some $\tau_0 \in \{\pm 1\}$.

In either case, there exists a sign $\sigma \in \{\pm 1\}$ for which the random variables

$$Z_n := \sum_{i > m_n} (b_i^{(n)} - \sigma a_i^{(n)}) \xi_{n,i}$$

converge to 0 in distribution. Since the coefficients are diffuse, Lemma 2.5 gives

$$\|b_{\text{tail}}^{(n)} - \sigma a_{\text{tail}}^{(n)}\|_2 \to 0.$$

Together with Lemma 3.4 this implies $\|b^{(n)} - \sigma a^{(n)}\|_2 \to 0$, contradicting

$$\|b^{(n)} - \sigma a^{(n)}\|_2^2 = \|a^{(n)}\|_2^2 + \|b^{(n)}\|_2^2 - 2\sigma \langle a^{(n)}, b^{(n)} \rangle = 2.$$

If the second alternative of Lemma 3.4 holds, then η is Rademacher and the head limits are multiples of η alone. The same argument applies, since in that case the good coordinates disappear from the head and the tail limit still lies on Γ. □

3.5. **A Gaussian alternative in the degenerate-head regime.**

The previous endpoint used only the support structure of infinitely divisible limits. The discussion below records a second route, showing that in the degenerate-head regime the diffuse tail automatically generates a nontrivial Gaussian component.

We start with a result that ensures that, in spite of the failure of the Lindeberg condition for the Central Limit Theorem, one may still ensure the existence of a non-degenerated Gaussian component of the limiting distribution.

Proposition 3.7 (Diffuse tails force a nontrivial Gaussian part)**.** *Let*

$$V_{n,i} := (a_i^{(n)} \xi_{n,i},\, b_i^{(n)} \xi_{n,i}) \in \mathbb{R}^2 \qquad (i > m_n),$$

where the coordinates are independent, centered, L^2-normalized, and satisfy $\|\xi_{n,i}\|_{L^1} \geq \delta$. Set

$$\alpha_{n,i} := \big((a_i^{(n)})^2 + (b_i^{(n)})^2\big)^{1/2}.$$

Assume

$$\max_{i > m_n} \alpha_{n,i} \to 0, \qquad \sum_{i > m_n} \alpha_{n,i}^2 \to \sigma^2 > 0,$$

and that the row sums $\sum_{i>m_n} V_{n,i}$ converge in distribution to an infinitely divisible law μ on $\mathbb{R}^2$. Then the covariance matrix of the Gaussian component of μ is not identically zero.

Sketch of proof. Set $\beta_n = \max_{i>m_n} \alpha_{n,i}$ and $r_n = \beta_n^{1/2}$. For each $i > m_n$ with $\alpha_{n,i} \neq 0$, let $K_{n,i} = r_n/\alpha_{n,i} \to \infty$ uniformly and define the centered truncation

$$\widetilde{\xi}_{n,i} := \xi_{n,i}\mathbf{1}_{\{|\xi_{n,i}|\le K_{n,i}\}} - \mathbb{E}[\xi_{n,i}\mathbf{1}_{\{|\xi_{n,i}|\le K_{n,i}\}}].$$

Since $\mathbb{E}[|\xi_{n,i}|] \ge \delta$ and $\mathbb{E}[\xi_{n,i}^2] = 1$, a simple estimate shows that there exists $c_0 = c_0(\delta) > 0$ such that

$$\mathbb{E}[\widetilde{\xi}_{n,i}^2] \ge c_0$$

for all large n and all $i > m_n$. Define $\widetilde{V}_{n,i} = (a_i^{(n)}\widetilde{\xi}_{n,i},\, b_i^{(n)}\widetilde{\xi}_{n,i})$. Then $|\widetilde{V}_{n,i}| \le 2r_n$, so the truncated array is bounded and infinitesimal, and

$$\sum_{i>m_n} \mathbb{E}[|\widetilde{V}_{n,i}|^2] \ge c_0 \sum_{i>m_n} \alpha_{n,i}^2 \longrightarrow c_0\sigma^2 > 0.$$

For infinitesimal triangular arrays, the covariance matrices of shrinking centered truncations converge to the Gaussian covariance of the limit law; see [**?**, Chapter IV]. Since the traces of the truncation covariances stay bounded away from zero, the Gaussian covariance of μ is nonzero. □

We now show an alternative route to the conclusion of the compactness proof of Theorem 1.1. Assume that we are in the degenerate-head regime $H_X \equiv H_Y \equiv 0$ – as the other one is essentially trivial – so the tail component μ, the law of (R_X, R_Y), is supported on $\Gamma = \{(u, v) : |u| = |v|\}$. Write $\mu = \gamma * \nu$, where γ is the Gaussian part and ν the independent remainder term. By Proposition 3.7, γ is nontrivial. Its support is a nonzero linear subspace $E \subset \mathbb{R}^2$.

Because $\operatorname{supp}\mu = \operatorname{supp}\gamma + \operatorname{supp}\nu \subset \Gamma = L_+ \cup L_-$, the connected set E must itself lie in one of the lines L_τ. Fix such a τ. For any $z \in \operatorname{supp}\nu$, the translated line $E + z$ is contained in Γ. Since a translate of L_τ not equal to L_τ intersects $L_+ \cup L_-$ in at most two points, one must have $z \in L_\tau$. Hence $\operatorname{supp}\nu \subset L_\tau$ and therefore $\operatorname{supp}\mu \subset L_\tau$. Thus $R_Y = \tau R_X$ almost surely, and hence this recovers the final contradiction in the first proof of Theorem 1.1.

4. Second Proof: Anticoncentration

This section gives a second proof of Theorem 1.1. The argument is quantitative in spirit. Instead of passing to a limit law, we split the coefficients into a head and a tail and show directly that one of two explicit mechanisms forces a positive modulus gap.

By Proposition 2.1, it is enough to prove an orthogonal lower bound for arbitrary unit vectors in the one-exception span. Thus let $a = (a_i)_{i\ge1}, b = (b_i)_{i\ge1} \in \ell^2(\mathbb{N})$ satisfy

$$\|a\|_2 = \|b\|_2 = 1, \qquad \langle a, b\rangle = 0,$$

and set

$$X := a_1\eta + \sum_{i\ge2} a_i\xi_i, \qquad Y := b_1\eta + \sum_{i\ge2} b_i\xi_i.$$

Because the family $(\eta, \xi_i)_{i\ge2}$ is orthonormal in L^2, these sums converge in L^2. We shall prove that

$$\||X| - |Y|\|_{L^2} \ge c_{\delta,\eta} > 0,$$

where the constant $c_{\delta,\eta}$ depends only on δ and on the law of η.

4.1. **The dichotomy setup.** We start with the basic setup for our second proof that is based on a simple dichotomy. Let

$$\varepsilon := \mathbb{E}[|X^2 - Y^2|], \qquad \Delta := \||X| - |Y|\|_{L^2},$$

and notice that

$$\varepsilon \le 2\Delta.$$

Fix

$$\kappa_{\delta,\eta} := \min\left(\frac{\delta}{4\sqrt{2}}, \frac{\mathbb{E}[|\eta - \widetilde{\eta}|]}{2\sqrt{2}}\right) > 0,$$

where $\widetilde{\eta}$ is an independent copy of η. Next set

$$\theta := \min\big(\delta^2/8, \delta\kappa_{\delta,\eta}\big), \qquad \lambda := \min\big(\delta^3/2^{20}, \delta\theta/2^{17}\big), \qquad s_0 := \frac{\theta^2}{2^{14}}.$$

Let S_η be the bounded linear probe from Lemma 3.3. Choose a constant $K_{\delta,\eta}$ so that

$$K_{\delta,\eta} \ge \max\left\{\frac{4}{\delta^2}, \|S_\eta\|_{L^\infty}\right\} \ge 1,$$

as $0 < \delta < 1$. In the case $\eta^2 \not\equiv 1$, let T_η be the bounded quadratic probe and increase $K_{\delta,\eta}$ (if necessary) to also satisfy

$$K_{\delta,\eta} \ge \|T_\eta\|_{L^\infty}.$$

Define the head set H by

$$H(a,b) := \{1\} \cup \{i \ge 2 : \max(|a_i|, |b_i|) > \lambda\}.$$

Then the cardinality of H satisfies

$$\#H \le 3 + 2\lambda^{-2} =: d^*(\delta,\eta),$$

and every tail coordinate $i \notin H$ satisfies

$$\max(|a_i|, |b_i|) \le \lambda.$$

The first step is to separate the proof into two mutually exclusive regimes. Either one of the two signs makes the tail nearly one-dimensional, in which case the argument is driven by head rigidity, or both signs retain substantial tail mass, in which case anticoncentration becomes effective. We gather this simple case split as the following result, whose proof we omit.

Lemma 4.1 (Tail sign dichotomy)**.** *Exactly one of the following alternatives holds:*

(i) *Diffuse: We have*

$$\|(a-b)\mathbf{1}_{H^c}\|_2^2 \ge 16 s_0 \quad \textit{and} \quad \|(a+b)\mathbf{1}_{H^c}\|_2^2 \ge 16 s_0.$$

(ii) *Concentrated: there exists $\tau \in \{\pm 1\}$ such that*

$$\|(a - \tau b)\mathbf{1}_{H^c}\|_2 < \frac{\theta}{32}.$$

4.2. **The concentrated branch.** We now turn to the first regime. In the concentrated case, the tail of $a - \tau b$ is small for a suitable sign τ, so any substantial discrepancy between the two vectors

must be visible on the head. The next proposition shows that the quadratic observable $X^2 - Y^2$ controls precisely those head coefficients.

Proposition 4.2 (Head entry bounds with one exceptional coordinate)**.** *The following estimates hold:*

(1) *For any indexes $i, j \in H \cap \{2, 3, \dots\}$,*

$$|a_i a_j - b_i b_j| \le K_{\delta,\eta}\, \varepsilon;$$

(2) *For any index $j \in H \cap \{2, 3, \dots\}$,*

$$|a_1 a_j - b_1 b_j| \le K_{\delta,\eta}\, \varepsilon;$$

(3) *if $\eta^2 \not\equiv 1$, then*

$$|a_1^2 - b_1^2| \le K_{\delta,\eta}\, \varepsilon.$$

Proof. For $i = j \ge 2$, use the quadratic probe T_i for ξ_i:

$$|a_i^2 - b_i^2| = \big|\mathbb{E}[(X^2 - Y^2)T_i]\big| \le \|T_i\|_{L^\infty}\varepsilon \le K_{\delta,\eta}\varepsilon.$$

For distinct $i, j \ge 2$, use the linear probes S_i, S_j:

$$2|a_i a_j - b_i b_j| = \big|\mathbb{E}[(X^2 - Y^2)S_i S_j]\big| \le \|S_i\|_{L^\infty}\|S_j\|_{L^\infty}\varepsilon \le K_{\delta,\eta}\varepsilon.$$

This proves (1).

For (2), use the product $S_\eta S_j$. Exactly as in the proof of Lemma 3.4, all terms vanish except the mixed exceptional-good term, so

$$2|a_1 a_j - b_1 b_j| = \big|\mathbb{E}[(X^2 - Y^2)S_\eta S_j]\big| \le \|S_\eta\|_{L^\infty}\|S_j\|_{L^\infty}\varepsilon \le K_{\delta,\eta}\varepsilon.$$

Finally, if $\eta^2 \not\equiv 1$, use T_η:

$$|a_1^2 - b_1^2| = \big|\mathbb{E}[(X^2 - Y^2)T_\eta]\big| \le \|T_\eta\|_{L^\infty}\varepsilon \le K_{\delta,\eta}\varepsilon.$$

□

To convert these entrywise bounds into a global contradiction, we use a basic fact about rank-one matrices. Intuitively, it isolates the geometric fact that two orthogonal unit vectors cannot agree too well on the head unless the corresponding head difference matrix is small.

Lemma 4.3 (Rank-one inequality)**.** *For $x, y \in \mathbb{R}^m$, let $A(x,y)_{ij} = x_i x_j - y_i y_j$ be the difference matrix between x and y. Then*

$$\|x - y\|_2^2\, \|x + y\|_2^2 \le 2\|A(x,y)\|_F^2.$$

Proof. Writing $s = \|x\|_2^2$, $t = \|y\|_2^2$, and $c = \langle x, y\rangle$, the left-hand side equals $(s+t)^2 - 4c^2$, while the right-hand side is $2(s^2 + t^2 - 2c^2)$. Thus the right-hand side has an extra factor $(s-t)^2$, which is nonnegative. □

Combining the head-entry bounds with the rank-one inequality yields the concentrated branch. The proof splits off the genuinely degenerate Rademacher-only head, since only there can the exceptional coordinate fail to provide a quadratic probe.

Proposition 4.4 (Concentrated branch)**.** *If there exists $\tau \in \{\pm 1\}$ such that*

$$\|(a - \tau b)\mathbf{1}_{H^c}\|_2 \le \frac{\theta}{32},$$

then there exists a constant $c_{\mathrm{line}}(\delta, \eta) > 0$ such that

$$\varepsilon \ge c_{\mathrm{line}}(\delta, \eta).$$

Proof. We distinguish two cases.

Case 1: η is not Rademacher, or $H \cap \{2, 3, \dots\}$ is nonempty. In this case the head coefficients are rigidly linked by Proposition 4.2: either directly through the diagonal estimate for the exceptional coordinate or, in the Rademacher case, through a good head coordinate exactly as in the proof of Lemma 3.4. In particular there exists a sign, which must be the same τ, such that the full head difference matrix of $a\mathbf{1}_H$ and $\tau b\mathbf{1}_H$ has all entries bounded by $K_{\delta,\eta}\varepsilon$. Thus, applying Lemma 4.3 for

$$x := a\mathbf{1}_H, \qquad y := \tau b\mathbf{1}_H,$$

we obtain

$$\|x - y\|_2^2 \, \|x + y\|_2^2 \le 2\#H^2 K_{\delta,\eta}^2 \varepsilon^2.$$

Since $(a - \tau b)\mathbf{1}_{H^c}$ has norm at most $\theta/32$ and $\|a - \tau b\|_2^2 = 2$ by orthogonality between a and b, the head part satisfies

$$\|x - y\|_2^2 = 2 - \|(a - \tau b)\mathbf{1}_{H^c}\|_2^2 \ge 1.$$

Hence

$$\|x + y\|_2 \le \sqrt{2}(\#H)K_{\delta,\eta}\varepsilon \le \sqrt{2}d^*(\delta, \eta)K_{\delta,\eta}\varepsilon.$$

Set

$$\alpha := \|(a + \tau b)\mathbf{1}_H\|_2, \qquad \beta := \|(a - \tau b)\mathbf{1}_{H^c}\|_2.$$

Then

$$\alpha \le \sqrt{2}\, d^*(\delta, \eta)K_{\delta,\eta}\varepsilon, \qquad \beta \le \theta/32.$$

We now spell out the gluing step. Write

$$p := (a - \tau b)\mathbf{1}_H, \qquad q := (a - \tau b)\mathbf{1}_{H^c},$$

$$r := (a + \tau b)\mathbf{1}_H, \qquad s := (a + \tau b)\mathbf{1}_{H^c},$$

and let

$$P := p_1\eta + \sum_{i \in H \cap \{2,3,\dots\}} p_i\xi_i, \qquad Q := \sum_{i \notin H} q_i\xi_i,$$

$$R := r_1\eta + \sum_{i \in H \cap \{2,3,\dots\}} r_i\xi_i, \qquad S := \sum_{i \notin H} s_i\xi_i.$$

Then

$$X - \tau Y = P + Q, \qquad X + \tau Y = R + S,$$

so

$$X^2 - Y^2 = (X - \tau Y)(X + \tau Y) = (P + Q)(R + S).$$

Because the supports of p, r and q, s are disjoint, the pair (P, R) is independent of the pair (Q, S); in particular P and S are independent. Therefore

$$|PS| \le |X^2 - Y^2| + |PR| + |QS| + |QR|,$$

and hence

$$\mathbb{E}[|P|]\,\mathbb{E}[|S|] = \mathbb{E}[|PS|] \le \varepsilon + \mathbb{E}[|PR|] + \mathbb{E}[|QS|] + \mathbb{E}[|QR|].$$

By Cauchy–Schwarz,

$$\mathbb{E}[|PR|] \le \|P\|_{L^2}\|R\|_{L^2}, \qquad \mathbb{E}[|QS|] \le \|Q\|_{L^2}\|S\|_{L^2}, \qquad \mathbb{E}[|QR|] \le \|Q\|_{L^2}\|R\|_{L^2}.$$

Moreover,

$$\|P\|_{L^2} = \|p\|_2 = \sqrt{2-\beta^2}, \qquad \|R\|_{L^2} = \|r\|_2 = \alpha,$$
$$\|Q\|_{L^2} = \|q\|_2 = \beta, \qquad \|S\|_{L^2} = \|s\|_2 = \sqrt{2-\alpha^2}.$$

Assume for contradiction that $\varepsilon < c_{\mathrm{line}}(\delta,\eta)$, where $c_{\mathrm{line}}(\delta,\eta)$ will be chosen so small that

$$\sqrt{2}\, d^*(\delta,\eta) K_{\delta,\eta} c_{\mathrm{line}}(\delta,\eta) \le 1.$$

Then $\alpha \le 1$, and therefore

$$\|P\|_{L^2} \ge 1, \qquad \|S\|_{L^2} \ge 1, \qquad \|P\|_{L^2}, \|S\|_{L^2} \le \sqrt{2}.$$

Consequently

$$\mathbb{E}[|P|]\,\mathbb{E}[|S|] \le \varepsilon + \sqrt{2}\,\alpha + \sqrt{2}\,\beta + \alpha\beta.$$

To lower-bound the left-hand side, write

$$P = p_1\eta + G, \qquad G := \sum_{i \in H \cap \{2,3,\dots\}} p_i \xi_i.$$

Either $|p_1| \ge \|p\|_2/\sqrt{2}$ or $\|G\|_{L^2} \ge \|p\|_2/\sqrt{2}$. In the first case, using an independent copy $\widetilde{\eta}$ of η,

$$2\mathbb{E}[|P|] = \mathbb{E}\big[|p_1\eta + G| + |p_1\widetilde{\eta} + G|\big] \ge |p_1|\,\mathbb{E}[|\eta - \widetilde{\eta}|] \ge 2\kappa_{\delta,\eta}\|p\|_2.$$

In the second case, let G' be an independent copy of G. Then

$$2\mathbb{E}[|P|] = \mathbb{E}\big[|p_1\eta + G| + |p_1\eta + G'|\big] \ge \mathbb{E}[|G - G'|].$$

Now

$$G - G' = \sum_{i \in H \cap \{2,3,\dots\}} p_i(\xi_i - \xi_i'),$$

where (ξ_i') is an independent copy of (ξ_i). The variables $\xi_i - \xi_i'$ are independent and centered, satisfy

$$\|\xi_i - \xi_i'\|_{L^2} = \sqrt{2}, \qquad \|\xi_i - \xi_i'\|_{L^1} \ge \delta,$$

and therefore

$$\|\xi_i - \xi_i'\|_{L^1} \ge \frac{\delta}{\sqrt{2}}\|\xi_i - \xi_i'\|_{L^2}.$$

Applying Lemma 2.4,

$$\mathbb{E}[|G - G'|] \ge \frac{\delta/\sqrt{2}}{2\sqrt{2}}\Big(\sum_{i \in H \cap \{2,3,\dots\}} p_i^2 \|\xi_i - \xi_i'\|_{L^2}^2\Big)^{1/2} = \frac{\delta}{2}\|G\|_{L^2} \ge 2\kappa_{\delta,\eta}\|p\|_2.$$

In both cases we obtain

$$\mathbb{E}[|P|] \geq \kappa_{\delta,\eta}\|p\|_2 \geq \kappa_{\delta,\eta}.$$

Since S is supported on H^c, only good coordinates enter it. Thus Lemma 2.4 gives

$$\mathbb{E}[|S|] \geq \frac{\delta}{2\sqrt{2}}\|s\|_2 \geq \frac{\delta}{2\sqrt{2}}.$$

Combining the last two displays,

$$\frac{\delta\kappa_{\delta,\eta}}{2\sqrt{2}} \leq \varepsilon + \sqrt{2}\,\alpha + \sqrt{2}\,\beta + \alpha\beta.$$

Using the bounds for α and β, we infer that

$$\frac{\delta\kappa_{\delta,\eta}}{2\sqrt{2}} \leq \Big(1 + 2d^*(\delta,\eta)K_{\delta,\eta} + \frac{\theta}{16}d^*(\delta,\eta)K_{\delta,\eta}\Big)\varepsilon + \frac{\sqrt{2}\,\theta}{32}.$$

By the choice of θ,

$$\frac{\sqrt{2}\,\theta}{32} \leq \frac{\delta\kappa_{\delta,\eta}}{8\sqrt{2}}.$$

Hence, after decreasing $c_{\text{line}}(\delta,\eta)$ if necessary so that

$$\Big(1 + 2d^*(\delta,\eta)K_{\delta,\eta} + \frac{\theta}{16}d^*(\delta,\eta)K_{\delta,\eta}\Big)c_{\text{line}}(\delta,\eta) \leq \frac{\delta\kappa_{\delta,\eta}}{8\sqrt{2}},$$

the right-hand side above is strictly smaller than $\delta\kappa_{\delta,\eta}/(2\sqrt{2})$ whenever $\varepsilon < c_{\text{line}}(\delta,\eta)$, a contradiction. Therefore $\varepsilon \geq c_{\text{line}}(\delta,\eta)$.

Case 2: η is Rademacher and $H = \{1\}$. Write

$$u := a\mathbf{1}_{H^c}, \qquad v := b\mathbf{1}_{H^c}, \qquad \rho := \theta/32,$$

and let

$$U := \sum_{i\geq 2} a_i\xi_i, \qquad V := \sum_{i\geq 2} b_i\xi_i.$$

Since $H = \{1\}$, every coordinate entering U and V is good. The concentrated assumption states precisely that

$$\|u - \tau v\|_2 = \|U - \tau V\|_{L^2} \leq \rho.$$

Set

$$R := U - \tau V, \qquad S := U + \tau V, \qquad \alpha := a_1 - \tau b_1, \qquad \beta := a_1 + \tau b_1.$$

Then

$$X - \tau Y = \alpha\eta + R, \qquad X + \tau Y = \beta\eta + S.$$

Because $\|a\|_2 = \|b\|_2 = 1$ and $\langle a, b\rangle = 0$, we have

$$\|a - \tau b\|_2^2 = \|a + \tau b\|_2^2 = 2.$$

Therefore

$$\alpha^2 + \|R\|_{L^2}^2 = \alpha^2 + \|u - \tau v\|_2^2 = 2, \qquad \beta^2 + \|S\|_{L^2}^2 = \beta^2 + \|u + \tau v\|_2^2 = 2.$$

In particular,

$$|\alpha| \geq \sqrt{2 - \rho^2} \geq 1.$$

Next, we estimate β. To this end, notice that

$$|\beta| \le |\alpha\beta| = |a_1^2 - b_1^2| = \big|\|v\|_2^2 - \|u\|_2^2\big| = \big|\langle u - \tau v, u + \tau v\rangle\big|,$$

thus it follows that

$$|\beta| \le \|u - \tau v\|_2 \|u + \tau v\|_2 \le \sqrt{2}\,\rho.$$

Hence $\|S\|_{L^2}^2 = \|u + \tau v\|_2^2 \ge 1$, and Lemma 2.4 gives

$$\|S\|_{L^1} = \Big\|\sum_{i\ge 2}(a_i + \tau b_i)\xi_i\Big\|_{L^1} \ge \frac{\delta}{2\sqrt{2}}\|u + \tau v\|_2 \ge \frac{\delta}{2\sqrt{2}}.$$

On the other hand,

$$\|R\|_{L^1} \le \|R\|_{L^2} \le \rho.$$

Now condition on the sigma-algebra generated by the good coordinates. Since η is an independent Rademacher variable,

$$X^2 - Y^2 = (X - \tau Y)(X + \tau Y) = (\alpha\eta + R)(\beta\eta + S) = \alpha\beta + RS + \eta(\alpha S + \beta R).$$

Therefore

$$\begin{aligned}\mathbb{E}\big[|X^2 - Y^2| \mid R, S\big] &= \frac{1}{2}\Big(|\alpha\beta + RS + \alpha S + \beta R| + |\alpha\beta + RS - \alpha S - \beta R|\Big)\\ &= \max\big(|\alpha\beta + RS|, |\alpha S + \beta R|\big) \ge |\alpha S + \beta R|.\end{aligned}$$

Taking expectations on both sides, by the law of iterated expectation we have

$$\mathbb{E}[|X^2 - Y^2|] \ge \|\alpha S + \beta R\|_{L^1} \ge |\alpha|\|S\|_{L^1} - |\beta|\|R\|_{L^1}.$$

Recalling the definition of ε and the estimates for the L^1 norm of S and R, we conclude that

$$\varepsilon \ge |\alpha|\|S\|_{L^1} - |\beta|\|R\|_{L^1} \ge \frac{\delta}{2\sqrt{2}} - \sqrt{2}\,\rho^2.$$

Since $\rho = \theta/32 \le \delta^2/256$ and $0 < \delta \le 1$, the right-hand side is bounded below by a positive constant depending only on δ (for instance by $\delta/4$). After decreasing $c_{\text{line}}(\delta, \eta)$ if necessary, this proves Case 2. □

4.3. **The diffuse branch.** We now move to the complementary regime. Here both $a - b$ and $a + b$ retain a definite amount of ℓ^2 mass on the tail, and every tail coordinate is individually small. This is exactly the setting in which block anticoncentration can be used.

Set

$$c := \frac{(a - b)\mathbf{1}_{H^c}}{\sqrt{2}}, \qquad d := \frac{(a + b)\mathbf{1}_{H^c}}{\sqrt{2}}.$$

Because $1 \in H$, the vectors c and d are supported only on good coordinates. In the diffuse case both have ℓ^2 norm at least $\sqrt{8s_0}$, while each coordinate is bounded by $\sqrt{2}\,\lambda$.

The next lemma provides the quantitative probabilistic input. It says that after arbitrary deterministic shifts, the two tail combinations cannot both remain close to zero with high probability.

Lemma 4.5 (Quantitative cross anticoncentration)**.** *There exists $r_* = r_*(\delta, \eta) > 0$ such that, in the diffuse case,*

$$\mathbb{P}\big(\min(|X_c + u|, |X_d + v|) > r_*\big) \ge \frac{1}{4}, \qquad (u, v \in \mathbb{R}).$$

Proof. We prove first a one-variable small-ball estimate. Let $e \in \ell^2(\mathbb{N})$ be supported on H^c that satisfies

$$\|e\|_2^2 \geq 8s_0 \qquad \text{and} \qquad \sup_{i\geq 2}|e_i| \leq \sqrt{2}\,\lambda.$$

We claim that there exists $r_0 = r_0(\delta,\eta) > 0$ such that

$$\mathbb{P}\big(|X_e - t| \leq r_0\big) \leq \frac{3}{8} \qquad (t \in \mathbb{R}). \tag{4.1}$$

Once this is proved, the lemma follows immediately by applying (4.1) to $e = c$ with $t = -u$ and to $e = d$ with $t = -v$. Indeed, in the diffuse case both c and d satisfy the displayed hypotheses, so

$$\mathbb{P}\big(|X_c + u| \leq r_0\big) \leq \frac{3}{8}, \qquad \mathbb{P}\big(|X_d + v| \leq r_0\big) \leq \frac{3}{8}.$$

By the union bound,

$$\mathbb{P}\big(|X_c + u| \leq r_0 \text{ or } |X_d + v| \leq r_0\big) \leq \frac{3}{4},$$

and therefore

$$\mathbb{P}\big(\min(|X_c + u|, |X_d + v|) > r_0\big) \geq \frac{1}{4}.$$

Thus it remains to establish the claim (4.1).

Step 1: a block partition. Set

$$M := \left\lceil 2^{16}\delta^{-2}\right\rceil, \qquad \eta := \frac{s_0}{2M}, \qquad r_0 := \frac{\delta\sqrt{\eta}}{16\sqrt{2}}.$$

Because $\delta \leq 1$ implies

$$M \leq 2^{17}\delta^{-2},$$

and because

$$2\lambda^2 \leq \frac{\delta^2\theta^2}{2^{33}} \leq \frac{\theta^2}{2^{15}M} = \eta,$$

every coordinate satisfies $e_i^2 \leq \eta$. Since

$$\|e\|_2^2 \geq 8s_0 > 2M\eta = s_0,$$

we may choose pairwise disjoint finite sets $G_1, \ldots, G_M \subset H^c$ such that

$$\eta \leq \sum_{i\in G_j} e_i^2 \leq 2\eta \qquad (1 \leq j \leq M). \tag{4.2}$$

Indeed, construct the blocks greedily. Suppose $G_1, \ldots, G_{j-1}$ have already been chosen. Their total energy is at most $2(j-1)\eta$, so the unused part of e still has energy at least

$$8s_0 - 2(j-1)\eta \geq 8s_0 - 2M\eta = 7s_0 > \eta.$$

Hence one can choose a finite subset of the remaining support whose energy is at least η, and by minimality its energy is at most $\eta + \sup_i e_i^2 \leq 2\eta$. This produces the next block.

Step 2: symmetrized block sums have uniform two-sided mass. Let X_e' be an independent copy of X_e, realized on a second copy of the probability space. For each $i \geq 2$ define

$$D_i := \xi_i(\omega_1) - \xi_i'(\omega_2).$$

Then the variables $(D_i)_{i\geq 2}$ are independent, centered, and symmetric, with

$$\|D_i\|_{L^2}^2 = \mathbb{E}[(\xi_i - \xi_i')^2] = 2.$$

Moreover, by Jensen's inequality applied conditionally on ξ_i,

$$\mathbb{E}[|D_i|] = \mathbb{E}\big[\mathbb{E}[|\xi_i - \xi_i'| \mid \xi_i]\big] \geq \mathbb{E}\big[|\xi_i - \mathbb{E}[\xi_i']|\big] = \mathbb{E}[|\xi_i|] \geq \delta.$$

Hence

$$\mathbb{E}[|D_i|] \geq \frac{\delta}{\sqrt{2}}\|D_i\|_{L^2}.$$

For each block define

$$Y_j := \sum_{i\in G_j} e_i D_i, \qquad \sigma_j^2 := \sum_{i\in G_j} e_i^2.$$

By (4.2),

$$\eta \leq \sigma_j^2 \leq 2\eta.$$

Applying Lemma 2.4 to the independent centered family $(D_i)_{i\in G_j}$, with lower L^1 ratio $\delta/\sqrt{2}$, gives

$$\mathbb{E}[|Y_j|] \geq \frac{(\delta/\sqrt{2})}{2\sqrt{2}}\Big(\sum_{i\in G_j} e_i^2\|D_i\|_{L^2}^2\Big)^{1/2} = \frac{\delta}{4}\big(2\sigma_j^2\big)^{1/2} = \frac{\delta}{2\sqrt{2}}\sigma_j \geq \frac{\delta\sqrt{\eta}}{2\sqrt{2}}.$$

Set

$$a_0 := \frac{\delta\sqrt{\eta}}{4\sqrt{2}} = 4r_0,$$

and notice that $a_0 \leq \mathbb{E}[|Y_j|]/2$. We define the event

$$A_j := \{|Y_j| \geq a_0\}.$$

Since

$$\mathbb{E}[Y_j^2] = 2\sigma_j^2 \leq 4\eta,$$

we obtain

$$\mathbb{E}[|Y_j|] \leq a_0 + \mathbb{E}[|Y_j|\mathbf{1}_{A_j}] \leq a_0 + \mathbb{P}(A_j)^{1/2}\mathbb{E}[Y_j^2]^{1/2} \leq a_0 + \mathbb{P}(A_j)^{1/2}(4\eta)^{1/2}.$$

Because $\mathbb{E}[|Y_j|] \geq 2a_0$, the second term on the right-hand side satisfies

$$\mathbb{P}(A_j)^{1/2}\,(4\eta)^{1/2} \geq a_0.$$

Therefore

$$\mathbb{P}(A_j) \geq \frac{a_0^2}{4\eta} = \frac{\delta^2}{128}. \tag{4.3}$$

Step 3: many active blocks with high probability. Let

$$N := \sum_{j=1}^{M} \mathbf{1}_{A_j},$$

and notice that it is a sum of independent random variables because the blocks are disjoint. By (4.3),

$$\mathbb{E}[N] = \sum_{j=1}^{M} \mathbb{P}(A_j) \geq M\frac{\delta^2}{128} \geq 512.$$

Since $\mathrm{Var}(\mathbf{1}_{A_j}) \le \mathbb{E}[\mathbf{1}_{A_j}]$, the independence implies that

$$\mathrm{Var}(N) = \sum_{j=1}^{M} \mathrm{Var}(\mathbf{1}_{A_j}) \le \sum_{j=1}^{M} \mathbb{E}[\mathbf{1}_{A_j}] = \mathbb{E}[N].$$

Chebyshev's inequality gives

$$\mathbb{P}(N < 256) \le \mathbb{P}\big(|N - \mathbb{E}[N]| \ge \mathbb{E}[N] - 256\big) \le \frac{\mathrm{Var}(N)}{(\mathbb{E}[N] - 256)^2} \le \frac{\mathbb{E}[N]}{(\mathbb{E}[N] - 256)^2} \le \frac{512}{256^2} < \frac{1}{16}.$$

Step 4: a Sperner bound for the symmetrized sum. Let

$$U := \sum_{j=1}^{M} Y_j.$$

We claim that for every $s \in \mathbb{R}$,

$$\mathbb{P}(|U - s| \le 2r_0) \le \frac{1}{8}. \tag{4.4}$$

Indeed,

$$\mathbb{P}(|U - s| \le 2r_0) \le \mathbb{P}(|U - s| \le 2r_0,\ N \ge 256) + \mathbb{P}(N < 256).$$

We have already bounded the second term by $1/16$, so it remains to estimate the first one.

Condition on the absolute values $(|Y_j|)_{j=1}^{M}$. Since each Y_j is symmetric, conditional on $(|Y_j|)_j$ the signs of the nonzero Y_j are independent Rademacher variables. On the event $\{N \ge 256\}$ let

$$J := \{j : |Y_j| \ge a_0\}, \qquad m := \#J \ge 256.$$

After freezing the inactive coordinates and the magnitudes $|Y_j|$, the condition $|U - s| \le 2r_0$ becomes

$$\sum_{j \in J} \varepsilon_j w_j \in I,$$

where $w_j := |Y_j| \ge a_0$ and I is an interval of diameter $4r_0 = a_0$. Since $a_0 < 2w_j$ for every $j \in J$, the family of sign patterns for which the weighted sum falls in I is an antichain in $\{\pm 1\}^J$. Indeed, if two such sign patterns were comparable in the natural subset order, their sums would differ by at least $2w_j \ge 2a_0 > a_0$, which is impossible inside an interval of diameter a_0. By Sperner's theorem, the conditional probability of this event is therefore at most

$$\frac{\binom{m}{\lfloor m/2 \rfloor}}{2^m} \le \frac{1}{\sqrt{m}} \le \frac{1}{16},$$

because $m \ge 256$. Averaging over the conditioning yields

$$\mathbb{P}(|U - s| \le 2r_0,\ N \ge 256) \le \frac{1}{16}.$$

Together with the bound for $\mathbb{P}(N < 256)$, this proves (4.4).

Step 5: unsymmetrization. Let

$$R := \sum_{i \notin G_1 \cup \cdots \cup G_M} e_i D_i,$$

and notice that R is independent of U, because it is built from coordinates disjoint from those entering the block sums. Also by construction, we have

$$X_e(\omega_1) - X'_e(\omega_2) = U + R.$$

Fix $t \in \mathbb{R}$. If both $|X_e(\omega_1) - t| \le r_0$ and $|X'_e(\omega_2) - t| \le r_0$, then

$$|X_e(\omega_1) - X'_e(\omega_2)| \le 2r_0.$$

Hence since X'_e is an independent copy of X_e,

$$\mathbb{P}(|X_e - t| \le r_0)^2 = \mathbb{P}(|X_e - t| \le r_0 \cap |X'_e - t| \le r_0) \le \mathbb{P}(|X_e(\omega_1) - X_e(\omega_2)| \le 2r_0) = \mathbb{P}(|U + R| \le 2r_0).$$

Conditioning on R and using (4.4),

$$\mathbb{P}(|U + R| \le 2r_0) \le \sup_{s \in \mathbb{R}} \mathbb{P}(|U - s| \le 2r_0) \le \frac{1}{8}.$$

Therefore

$$\mathbb{P}(|X_e - t| \le r_0) \le \frac{1}{\sqrt{8}} < \frac{3}{8}.$$

This proves (4.1), and with it the lemma. □

Once this shifted anticoncentration estimate is available, the lower bound for ε follows from the same factorization of $X^2 - Y^2$ that was already used in the compactness proof.

Proposition 4.6 (Diffuse branch)**.** *If the diffuse alternative of Lemma 4.1 holds, then*

$$\varepsilon \ge c_{\text{cross}}(\delta, \eta) > 0.$$

Proof. Let

$$h_\pm := X_{(a \pm b)\mathbf{1}_H}.$$

Then

$$X^2 - Y^2 = 2\big(X_d + h_+/\sqrt{2}\big)\big(X_c + h_-/\sqrt{2}\big),$$

and therefore

$$|X^2 - Y^2| \ge 2 \min\Big(|X_d + h_+/\sqrt{2}|, |X_c + h_-/\sqrt{2}|\Big)^2.$$

Condition on the head variables $h_\pm$ and apply Lemma 4.5 with the shifts $u = h_-/\sqrt{2}$ and $v = h_+/\sqrt{2}$. This yields a uniform positive lower bound for the conditional expectation of the squared minimum, hence for ε. The resulting constant depends only on δ and η. □

4.4. **Conclusion of the second proof.** At this point the proof is immediate: the dichotomy shows that every orthogonal pair falls into exactly one of the two branches, and each branch gives a quantitative lower bound for the same quantity ε.

Second proof of Theorem 1.1. By Proposition 2.1, it is enough to prove a uniform orthogonal lower bound. Let a, b be orthogonal unit vectors and form X and Y as above. Apply the dichotomy of Lemma 4.1: if the concentrated alternative holds, then Proposition 4.4 yields

$$\varepsilon \ge c_{\text{line}}(\delta, \eta).$$

If the diffuse alternative holds, then Proposition 4.6 yields

$$\varepsilon \geq c_{\text{cross}}(\delta, \eta).$$

Since $\varepsilon \leq 2\Delta$, we conclude that

$$\Delta = \||X| - |Y|\|_{L^2} \geq \frac{1}{2} \min\big(c_{\text{line}}(\delta, \eta), c_{\text{cross}}(\delta, \eta)\big) =: c_{\delta,\eta} > 0.$$

The orthogonal reduction then gives

$$\min_{\sigma \in \{\pm 1\}} \|X - \sigma Y\|_{L^2} \leq \frac{\sqrt{2}}{c_{\delta,\eta}} \||X| - |Y|\|_{L^2}$$

for all $X, Y \in \overline{\text{span}}(\eta, \xi_i : i \geq 2)$, completing the proof of Theorem 1.1. □

5. Comments and Open Problems

The compactness and quantitative proofs suggest several natural directions for further work. Although the theorem proved here settles the qualitative one-exception problem in the real Hilbertian setting, the mechanisms that enter the proof are flexible enough that one is led almost immediately to adjacent questions.

Question 5.1. Classical phase retrieval is recovery from the absolute value nonlinearity modulo the group $\{\pm 1\}$. It is natural to ask: are there analogous theorems for other nonlinearities modulo possibly larger compact symmetry groups?

The compactness proof is particularly well suited to the above direction, since it separates the problem into an exact limit identity and a geometric classification of the corresponding support set. It points toward a broader geometric framework in which the support theorem, rather than the specific real-sign symmetry, is the central object.

The next question concerns the extent to which the present theorem can be sharpened, and how much should persist in a wider Banach-function-space setting.

Question 5.2. Which Banach-function-space analogues of the present theorem remain true once the ambient space is no longer Hilbertian? In particular, is there an L^p-analogue of our result when $p < 2$? The answer is yes when $p \geq 2$, as an immediate consequence of the interpolation/extrapolation theory developed in [**?**].

This problem seems especially interesting, as the function-space formulation of stable phase retrieval is not inherently Hilbertian, even though the present proof is. Any successful extension would likely require replacing exact orthogonality by a more flexible geometric notion while retaining enough structure to control the dichotomy between near and far vectors.

Question 5.3. To what extent can independence be weakened in the above theorems?

The motivation for the above problem is that there are many examples of sequences of functions (for example, sparse Fourier series) that behave in certain senses like independent random variables. To what extent can the methods of this paper be generalized to cover these examples as well?